

Dell Technologies Case Competition

FusionTech Systems: Unlocking Customer Insights with AI

Note: This case is fictional and created for educational purposes only.

Your Challenge

Your team will step into the shoes of Dell Technologies' Consultants helping FusionTech Systems solve a real business problem using AI and Dell infrastructure. You'll analyze customer data, design an AI solution, and present your recommendations to FusionTech's executives.

What You'll Learn

- How to analyze real customer data to find business insights
- How AI/ML can solve actual business problems
- How technology infrastructure enables AI at scale
- How to present technical recommendations to business leaders
- What Dell Technologies does and how to prepare for interviews

 *This case is designed for ALL majors—whether you're in business, marketing, CS, MIS, finance, HR, or any other field, you'll find ways to contribute your unique perspective!*

About FusionTech Systems

FusionTech Systems is a retail technology company (think similar to Apple, Roku, Peloton, Whoop) that sells directly to consumers through online and in store channels. Here's their snapshot:

Annual Revenue	\$2.8 Billion
Employees	~15,000
Global Facilities	45 worldwide, for product development and distribution

The Problem

FusionTech is drowning in data. As an direct-to-consumer technology company, their data has grown by 300% from customer reviews, product searches, support tickets, website analytics, and inventory systems. But their datacenter technology infrastructure is 8 years old and can't keep up:

- **System outages and crashes** that disrupt operations
- **Multi-day delays** to get basic analytics reports
- **Stalled AI/ML projects** costing \$2.4M per quarter in lost opportunities

 **Leadership's Mandate:** FusionTech needs to modernize their datacenter infrastructure in 12 months AND launch one high-impact AI solution that improves customer experience.

Three Business Challenges (Pick One)

FusionTech has identified three critical problems hurting their business. Your team will choose ONE to solve with AI:

Challenge 1: Customer Feedback Blind Spots

The Problem: FusionTech collects thousands of product reviews from customers who buy their products through their FusionTech.com platform. But they can't analyze them fast enough. Merchandising and buying teams wait several days for manual summaries, which delays decisions about which products to promote, discontinue, or discount.

Why It Matters: If customers are complaining that a specific laptop model overheats or slows down with typical use, FusionTech needs to know immediately. By the time they read manual reports days later, they've already featured that problematic product in their weekly email to 2 million customers, leading to returns, negative reviews, and loss of trust.

Challenge 2: One-Size-Fits-All Marketing

The Problem: FusionTech has diverse customers, from gamers who need high-performance PCs, to business professionals who want reliable workstations, or new parents needing a laptops that can survive drops and spills. Yet they send the same promotional emails and show the same homepage to everyone. Campaign effectiveness and conversion rates are declining.

Why It Matters: Imagine a professional photographer getting ads for a budget laptop, or a gamer seeing promotions for business workstations. Generic marketing wastes money and annoys customers. Personalized product recommendations and targeted campaigns could increase conversion rates and reduce unsubscribe rates.

Challenge 3: Overwhelmed Customer Support

The Problem: Customer support requests keep increasing. Long wait times, inconsistent answers from different agents, and frustrated customers are hurting satisfaction scores. The old ticketing system can't handle the volume.

Why It Matters: When customers can't get help quickly, they share bad reviews, switch to competitors, and hurt FusionTech's reputation. Better support = happier customers = more repeat business.

Your Dataset: FusionTech Customer Reviews

File: Attached via email

What's Inside: A snapshot of real customer product ratings and detailed written reviews of electronics and tech products sold through the FusionTech.com platform, along with product information and reviewer details. This data represents the authentic voice of FusionTech's customers shopping for technology.

Why This Dataset Matters:

- **Find the biggest customer pain points** (e.g., battery life, overheating, broken hinges, slow support)
- **Understand what customers really care about** in their own words
- **Justify your AI solution** with data-driven evidence
- **Show FusionTech's executives that you can work with real, messy data**

 **Data Reality Check:** This dataset has multilingual text (mostly English with some Spanish), missing values, duplicates, and repeated product specs, just like real-world data! You will need to clean it before analyzing.

 **Pro Tip:** You can analyze a subset of the data (e.g., last 12 months or one product line) if you explain your reasoning. Focus on quality insights, not analyzing everything.

What You'll Build

Part 1: Design Your AI/ML Solution

Create an end-to-end AI solution for your chosen challenge. Think of this like a blueprint for how FusionTech would actually implement this:

a) Data Preparation

Explain how you'd prepare the data for your AI model. What data sources do you need? How would you clean and organize it?

Example: "We'd combine customer reviews from the CSV with support ticket data, clean up duplicates and missing values, and label reviews by product category."

b) Choose Your AI Approach

Select the AI/ML method that best solves your problem:

- **Sentiment Analysis:** Automatically detect if reviews are positive, negative, or neutral
- **Topic Modeling:** Find common themes in customer feedback (e.g., 'battery problems,' 'great design')
- **Customer Segmentation:** Group customers by behavior, needs, or preferences
- **Recommender Systems:** Suggest products based on customer preferences
- **Intent Classification:** Categorize support requests automatically
- **Chatbot/Agent Assist:** Help support agents answer questions faster

 **Pro Tip:** As a team, you have creative freedom in how you would like to present your AI solution. Just keep in mind that you will be sharing screen.

c) How It Works in Practice

Describe who uses your AI solution, where it shows up in their daily work, and how it improves a key metric.

Example: "Merchandising managers would see a daily dashboard showing the top 5 customer complaints from yesterday's reviews. This reduces the time to identify problematic products from 5 days to 1 day, helping them adjust product placement and promotions 80% faster."

d) Responsible AI Considerations

Address the ethical and practical concerns of your AI solution:

- **Privacy:** How do you protect customer data?
- **Bias:** Could your AI treat some customer groups unfairly?
- **Human-in-the-Loop:** When should humans review AI decisions?

Part 2: Recommend Dell Technologies Infrastructure

Remember, FusionTech's 8-year-old infrastructure is a bottleneck that's stalling AI projects. Your AI solution needs modern infrastructure to work reliably and scale.

Choose one main Dell Technologies product that addresses FusionTech's infrastructure problems AND supports your AI workload:

Product	Best For
PowerScale	Massive unstructured data (images, videos, logs, documents) that AI needs to process
PowerStore	Fast, flexible storage for databases and apps that need quick AI responses
PowerEdge	High-performance computing infrastructure optimized for AI workloads, machine learning training, and data-intensive processing with GPU acceleration
PowerProtect	Data backup and recovery to protect against data loss

Your recommendation should explain:

1. **How it supports your specific AI solution** (e.g., "PowerScale handles the millions of customer review text files our sentiment analysis needs")
2. **How it fixes FusionTech's infrastructure problems** (e.g., eliminates outages, speeds up analytics)
3. **Why this product vs. an alternative** (compare to at least one other Dell product)

Your Presentation: What to Include

Format: 10-minute presentation to FusionTech executives

Audience: Executive team with diverse backgrounds (finance, operations, technology, marketing)

Requirement: All team members must speak during the presentation

Required Sections

1. Executive Summary (2 minutes)

- Which challenge you chose and why
- Your recommended AI solution in one sentence
- Top 2-3 insights from the customer review data
- Your Dell Technologies infrastructure recommendation

2. Data Analysis (2 minutes)

- How you cleaned the data (briefly)
- Key insights: sentiment trends, recurring themes, biggest pain points
- Visual charts: graphs, word clouds, or topic summaries

3. AI/ML Solution (2 minutes)

- What type of AI/ML method you're using (e.g., sentiment analysis, clustering)
- Why this method fits your business challenge
- Responsible AI: privacy, bias mitigation, human oversight

4. Dell Technologies Infrastructure (2 minutes)

- Your chosen Dell product (e.g., PowerScale, PowerStore, PowerEdge, etc.)
- How it enables your AI solution
- How it solves FusionTech's infrastructure bottlenecks
- Why you chose this product vs. an alternative (compare to at least one other option)

5. Business Impact & Implementation (1.5 minutes)

- Which metrics improve (e.g., customer satisfaction, response time, marketing ROI)
- Expected impact over 12 months
- Simple timeline: 30 days → 90 days → 12 months

6. Risk Assessment & Mitigation (0.5 minutes)

Identify 2-3 biggest risks and how you'd address them:

- Data quality issues
- AI misinterpretation
- Privacy concerns
- Employee adoption challenges

Presentation Tips

- **Keep it visual:** Use charts, diagrams, and images instead of bullet-heavy slides
- **Tell a story:** Walk executives through the problem → data insights → solution → impact
- **Be concise:** You have 10 minutes—practice timing!
- **Everyone speaks:** Divide content so all team members present
- **Focus on business value:** Executives care about ROI, not technical details

How You'll Be Evaluated

Criteria	Weight	What We're Looking For
Dataset Analysis	25%	Did you clean the data properly? Are your insights accurate & meaningful? Do your visuals tell a story?
AI/ML Solution	25%	Is your AI approach realistic and well-suited to the problem? Can you explain it clearly? Did you consider responsible AI?
Business Insights & Recommendations	20%	Are your recommendations practical? Do they align with FusionTech's needs? Is the expected business impact clear and believable?
Dell Infrastructure Recommendation	15%	Did you choose a Dell product for your AI workload? Can you explain why this product vs. alternatives? Does it solve FusionTech's infrastructure problems?
Presentation Quality	15%	Is your presentation clear, engaging, and well-organized? Do all team members speak? Are your slides professional and visual?

Timeline & Deadlines

Submission Deadline	Wednesday, April 1, 2026 by 2:00 PM EST / 1:00 PM CST
Presentation Dates	Thursday, April 2, 2026 (1-hour slot) Friday, April 3, 2026 (1-hour slot)

Resources to Help You Succeed

Learn About Dell Technologies

- [Dell.com/AI](https://dell.com/AI) - Explore AI and infrastructure solutions
- [Dell.com/Careers](https://dell.com/Careers) - Learn about roles and interview tips
- Dell Technologies YouTube - Watch product demos and customer stories

Data Analysis Tools

- Excel or Google Sheets - For basic data cleaning and charts
- Python (pandas, matplotlib) - For more advanced analysis (optional)
- Tableau Public or Power BI - For professional visualizations (optional)
- Word cloud generators - For visualizing common themes

Final Thoughts

This case is your opportunity to think like a business consultant, data analyst, and technology advisor all in one. You'll work with real data, design a real solution, and present to executives.

Remember:

- **Focus on the business problem**, not just the technology
- **Let the data guide your decisions**, not assumptions
- **Tell a compelling story** that connects insights to action
- **Show how Dell Technologies enables your vision**

Good luck! We can't wait to see what you create 🚀

Frequently Asked Questions

Q: Do I need to know how to code AI/ML models?

A: No! You need to understand what AI/ML can do and recommend the right approach, but you don't need to build the models yourself – this is optional and depends on your team's expertise.

Q: What if I'm a marketing/finance/business major and don't know much about technology?

A: Perfect! This case needs business thinkers. Focus on the customer insights, business impact, and storytelling—your technical teammates can help with the infrastructure recommendations.

Q: Can we use external data sources beyond the provided dataset?

A: Yes! You can mention how you'd combine the customer reviews with other realistic data sources (e.g., support tickets, sales data), but your PRIMARY analysis should be based on the provided dataset.

Q: How technical should our presentation be?

A: Strike a balance. Explain the technology clearly enough that a non-technical executive understands, but show enough depth that a technical executive knows you've done your homework. Avoid jargon—use analogies and examples.

Q: What if we can't decide on which challenge to pick?

A: Look at your dataset! Which challenge has the strongest evidence in the customer reviews? Which one does your team feel most passionate about? When in doubt, pick the one where you can tell the best story.

Q: How do we prepare for Dell interviews after this case?

A: You'll understand Dell's product portfolio, how infrastructure enables AI, and how to present business recommendations. Visit [Dell.com/Careers](https://www.dell.com/careers) to see what roles interest you, and be ready to discuss this case experience in interviews!